FAN-HHTT
Electrical System

Single Line Diagram (SLD)
Project Overview & Equipment Summary

Document Field	Details
Project	FAN-HHTT Electrical System
Document Type	Single Line Diagram (SLD)
Area	Top Deck, Switchgear Room 04BFA
Switchgear Panel	04BFA

Document content arranged across 10 pages for readability. This overview is based solely on the supplied text; it is not a substitute for the approved electrical drawing.

1. Document Overview

This document describes the low-voltage (LV) switchgear arrangement for feeders 04BFA-01 through 04BFA-08 within the FAN-HHTT Electrical System project.

Document identification

Attribute	Value
Project	FAN-HHTT Electrical System
Document type	Single Line Diagram (SLD)
Location / area	Top Deck, Switchgear Room 04BFA
Equipment designation	Switchgear Panel 04BFA

Purpose

The supplied description summarizes the LV switchgear lineup, incoming supply, bus coupler configuration, protective relays, and short-circuit withstand rating. Equipment ratings and design references are retained as provided.

Scope note

The source text names feeders 04BFA-01 through 04BFA-08, while the detailed equipment list specifies MCC feeders 04BFA-01 through 04BFA-04 only. Details for feeders 05–08 were not supplied and have not been inferred.

2. System Description

The SLD represents the LV switchgear arrangement associated with panel 04BFA. The incoming supply is identified as originating from transformer TR-02, rated 400 V and 50 Hz on the LV side.

Key system elements

Element	Supplied information
LV switchgear	Panel 04BFA
Incoming source	Transformer TR-02; 400 V, 50 Hz
Bus arrangement	Two incomers and a bus coupler
Bus coupler status	Normally Open (NO)
Protection functions	50/51 and 50N/51N
Short-circuit rating	50 kA for 1 second

Interpretation

The listed configuration indicates a switchgear system with two 1600 A incomers and a 1600 A normally open bus coupler. The exact operating philosophy, interlocking, and bus-section arrangement must be confirmed against the approved SLD.

3. Incoming Supply & Transformer

Transformer TR-02 is identified as the source transformer for the LV supply.

Parameter	Value
Transformer tag	TR-02
Rated power	1 MVA
Voltage ratio	11 kV / 0.4 kV
Vector group	Dyn11
LV nominal voltage	400 V
Frequency	50 Hz

Supply path (textual summary)

Upstream HV interface → Transformer TR-02 (11 kV / 0.4 kV) → LV switchgear panel 04BFA.

Verification note

The upstream HV interface should be checked against drawing RAN-HHTT-0000EL-SLD-0018. The source text does not provide transformer impedance, earthing details, cable information, or the detailed HV protection arrangement.

4. Incoming Circuit Breakers

Two incoming circuit breakers are listed for switchgear panel 04BFA.

Device	Rating / type	Protection
Incomer A	1600 A ACB	Micrologic protection
Incomer B	1600 A ACB	Micrologic protection

Equipment notes

• ACB means Air Circuit Breaker.
• Both incomers are specified at 1600 A.
• The revision history records an update to the incomer ACB model in Rev 04.
• The exact manufacturer, model number, trip settings, and interlocking logic are not included in the supplied text.

Design checks

Confirm the approved breaker model and settings, coordination with downstream MCCBs, short-circuit breaking capacity, and any incomer/bus-coupler interlocks from the controlled engineering documents.

Deep Space Mission Briefing â€” Page 1 of 50
Mission: Asteria 001
Objective: Conduct autonomous navigation trials beyond lunar orbit.
Payload: Micro-probe swarm with optical and infrared sensors.
Trajectory: Gravity assist via cislunar loop, then outbound to heliocentric drift.
Comms: Ka-band with adaptive beamforming; 15-minute cadence.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 002
Objective: Validate radiation-hardened avionics under solar storm conditions.
Payload: Fault-tolerant compute module; solar particle detector.
Trajectory: Sun-pointed drift with periodic roll maneuvers.
Comms: Store-and-forward; dynamic rate adaptation.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 003
Objective: Test cold-start procedures in deep-space thermal environments.
Payload: Phase-change thermal regulation demonstrator.
Trajectory: Heliocentric orbit with aphelion thermal dips.
Comms: Burst mode telemetry with error correction.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 004
Objective: Characterize micrometeoroid flux in interplanetary space.
Payload: Impact sensor array; debris spectrometer.
Trajectory: Slow spirals through inner heliosphere.
Comms: Scheduled housekeeping and science frames.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 005
Objective: Validate autonomous star tracker calibration.
Payload: Wide-field star tracker; onboard catalog subset.
Trajectory: Cruise with slew maneuvers for calibration.
Comms: Low-rate trickle plus event flags.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 006
Objective: Evaluate ion thruster long-duration cycling.
Payload: Miniature ion drive; propellant gauging sensor.
Trajectory: Shallow spiral to simulate station-keeping.
Comms: Engineering downlink with thrust telemetry.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 007
Objective: Demonstrate fault detection, isolation, and recovery (FDIR).
Payload: Redundant bus; watchdog supervisors.
Trajectory: Nominal cruise with injected anomalies.
Comms: Health frames and anomaly reports.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 008
Objective: Test adaptive sun sensor under variable incidence.
Payload: Multi-quadrant sun sensor.
Trajectory: Spin-stabilized segment with precession.
Comms: Beacon plus periodic sensor dumps.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 009
Objective: Verify deep-space time synchronization without GNSS.
Payload: Ultra-stable oscillator; pulsar timing experiment.
Trajectory: Deep-space drift.
Comms: Time-transfer exchanges with ground.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 010
Objective: Assess dust charging phenomena on exposed surfaces.
Payload: Electrometer arrays; dust collectors.
Trajectory: Sunward arcs with perihelion passes.
Comms: Science frames at perihelion.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 011
Objective: Evaluate autonomous fault-tolerant routing in probe swarm.
Payload: Inter-probe RF links; consensus algorithms.
Trajectory: Formation flight in loose cluster.
Comms: Crosslinks plus Earth relay.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 012
Objective: Radiation belt traversal endurance.
Payload: Dosimeters; latchup monitors.
Trajectory: Artificial belt analog via solar events.
Comms: Continuous housekeeping during peaks.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 013
Objective: Test optical beacon for interstellar navigation concepts.
Payload: Laser beacon; photodiode array.
Trajectory: Distant cruise with periodic flashes.
Comms: Beacon health and timing.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 014
Objective: Validate autonomous momentum management.
Payload: Reaction wheels; magnetorquers.
Trajectory: Slew sequences and wheel offloading.
Comms: Wheel telemetry and control states.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 015
Objective: Cold-gas thruster precision pointing trials.
Payload: Micro-thrusters; fine sun sensor.
Trajectory: Pointing holds targeting bright stars.
Comms: Attitude error logs and corrections.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 016
Objective: Solar sail material endurance test.
Payload: Sail patch; deployment rig.
Trajectory: Controlled illumination cycles.
Comms: Imagery plus tension telemetry.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 017
Objective: Validate star occultation sensing pipeline.
Payload: Photometer; edge-detection firmware.
Trajectory: Line-of-sight scans past bright objects.
Comms: Photometry frames and event flags.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 018
Objective: Thermoelastic deformation characterization.
Payload: Strain gauges; temperature probes.
Trajectory: Heating and cooling cycles.
Comms: Structural telemetry packets.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 019
Objective: Validate star camera lost-in-space recovery.
Payload: Star camera; pattern matcher.
Trajectory: Randomized attitude slews.
Comms: Recovery time metrics.
Notes: This content is fictional and unrelated to any existing materials in the source document.

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Mission: Asteria 020
Objective: Onboard compression efficiency trials.
Payload: FPGA compressor; sample datasets.
Trajectory: Not applicable.
Comms: Throughput and ratio reports.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 021
Objective: Antenna deployment reliability checks.
Payload: Tape-spring antenna; latch sensors.
Trajectory: Spin and non-spin deployments.
Comms: Antenna gain and health.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 022
Objective: Power system deep-discharge recovery.
Payload: Battery emulator; protection circuits.
Trajectory: Eclipse simulations.
Comms: Voltage and current logs.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 023
Objective: Star tracker glare rejection assessment.
Payload: Baffles; image processing filters.
Trajectory: Solar limb approaches.
Comms: Image snippets and stats.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 024
Objective: RF interference susceptibility testing.
Payload: EMI generators; spectrum monitoring.
Trajectory: Not applicable.
Comms: Spectral scans and alerts.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 025
Objective: Long-baseline clock drift characterization.
Payload: Oscillator comparison rig.
Trajectory: Long coasts.
Comms: Drift logs at intervals.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 026
Objective: Autonomy framework stress test.
Payload: Decision engine; fault injectors.
Trajectory: Scenario scripts in flight.
Comms: State transitions and outcomes.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 027
Objective: Optical comm pointing experiment (low-power).
Payload: MEMS mirror; beacon detector.
Trajectory: Fine pointing trials.
Comms: Link acquisition attempts.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 028
Objective: Sail attitude control via reflectivity modulation.
Payload: Variable albedo patches.
Trajectory: Low-thrust attitude maneuvers.
Comms: Attitude and illumination logs.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 029
Objective: Cold-soak electronics validation.
Payload: Thermal plates and sensors.
Trajectory: Shadowed cruise segments.
Comms: Thermal health and rewarm times.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 030
Objective: Autopilot gain scheduling under jitter.
Payload: Control firmware with tunable loops.
Trajectory: Induced jitter maneuvers.
Comms: Stability margins and logs.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 031
Objective: Long-range optical imaging of background stars.
Payload: Small telescope; CMOS imager.
Trajectory: Stabilized drift.
Comms: Compressed images at intervals.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 032
Objective: Deep-space dust plume sampling.
Payload: Aerogel collector; micro-spectrometer.
Trajectory: Plume intercept maneuvers.
Comms: Sample logs and spectra.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 033
Objective: Reaction wheel desaturation strategy trials.
Payload: Wheel set; magnetorquers.
Trajectory: Desat sequences under constraints.
Comms: Control effort metrics.
Notes: This content is fictional and unrelated to any existing materials in the source document.

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Mission: Asteria 034
Objective: Radiation-tolerant storage endurance.
Payload: MRAM; ECC monitors.
Trajectory: Extended exposure periods.
Comms: Error rates and corrections.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 035
Objective: Target tracking under low SNR conditions.
Payload: Kalman filter variants; test targets.
Trajectory: Simulated target passes.
Comms: Tracking performance summaries.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 036
Objective: Dust impact shock isolation test.
Payload: Isolation mounts; accelerometers.
Trajectory: Induced impact testing.
Comms: Vibration spectra and damping.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 037
Objective: Star camera pixel defect mapping.
Payload: Dark frames; hot pixel detection.
Trajectory: Dark sky imaging.
Comms: Defect maps and masks.
Notes: This content is fictional and unrelated to any existing materials in the source document.

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Mission: Asteria 038
Objective: Low-power hibernation wake-up tests.
Payload: Timed wake controllers.
Trajectory: Long-duration coasts.
Comms: Wake success rates and times.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 039
Objective: Autonomous sun-pointing safe mode evaluation.
Payload: Safe-mode controller.
Trajectory: Induced safing entries.
Comms: Entry/exit telemetry.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Deep Space Mission Briefing â€” Page 40 of 50
Mission: Asteria 040
Objective: Onboard event-driven compression.
Payload: Triggered capture framework.
Trajectory: Not applicable.
Comms: Event logs and compressed frames.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 041
Objective: Flexible power point tracking algorithm test.
Payload: Solar array emulator.
Trajectory: Varying illumination trials.
Comms: Power curves and efficiency.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 042
Objective: Micro-propulsion plume impingement study.
Payload: Plume sensor; witness plates.
Trajectory: Short burns near sensors.
Comms: Impingement measurements.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 043
Objective: Autonomy under delayed command uplinks.
Payload: Delay injectors; plan executor.
Trajectory: Nominal cruise with delayed ops.
Comms: Plan outcome reports.
Notes: This content is fictional and unrelated to any existing materials in the source document.

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Mission: Asteria 044
Objective: Star tracker blinding event response.
Payload: Glare sensors; fallback estimators.
Trajectory: Solar approaches.
Comms: Event flags and response times.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 045
Objective: Long-haul thermal radiator performance.
Payload: Variable emissivity panels.
Trajectory: Warm/cold cycling.
Comms: Heat rejection metrics.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 046
Objective: Redundant bus cross-strap validation.
Payload: Cross-strap matrix; health monitors.
Trajectory: Induced failures.
Comms: Switchover logs and continuity.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 047
Objective: Optical bench alignment stability.
Payload: Metrology targets; alignment sensors.
Trajectory: Vibration injection tests.
Comms: Alignment drifts over time.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Deep Space Mission Briefing â€” Page 48 of 50
Mission: Asteria 048
Objective: Autonomous fault tree evaluation.
Payload: Diagnostics engine; synthetic faults.
Trajectory: Scenario execution.
Comms: Decision traces and results.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 049
Objective: Radiation-hardened bus performance trending.
Payload: Monitors; A/B bus usage.
Trajectory: Extended operation.
Comms: Trending summaries.
Notes: This content is fictional and unrelated to any existing materials in the source document.

Mission: Asteria 050
Objective: End-to-end autonomy rehearsal.
Payload: Full mission stack.
Trajectory: Comprehensive scripted run.
Comms: Final outcome metrics.
Notes: This content is fictional and unrelated to any existing materials in the source document.

5. Bus Coupler & Bus Arrangement

The switchgear includes a 1600 A bus coupler ACB specified as Normally Open (NO).

Parameter	Supplied value
Device	Bus coupler
Breaker type	ACB
Rated current	1600 A
Normal status	Normally Open (NO)

Functional summary

A bus coupler can connect two bus sections when the approved operating conditions permit. The stated normally open status describes its normal configuration; it does not by itself define the transfer sequence or permitted switching conditions.

Important verification

Check the approved SLD and switching philosophy for bus-section boundaries, electrical/mechanical interlocks, source paralleling prevention, and the conditions under which the coupler may be closed.

6. Protection & Short-Circuit Rating

The source description identifies protection functions 50/51 and 50N/51N and a rated short-circuit capacity of 50 kA for 1 second.

ANSI device number	Common function
50	Instantaneous overcurrent protection
51	Time-delayed overcurrent protection
50N	Instantaneous earth/neutral overcurrent protection
51N	Time-delayed earth/neutral overcurrent protection

Short-circuit withstand

Rated value provided: 50 kA for 1 second. The relevant rating terminology and applicability to the complete assembly should be verified against the switchgear manufacturer's data and approved fault-level study.

Coordination note

Protective devices are stated to be coordinated in accordance with IEC 60947. Actual relay curves, pickup values, time delays, and discrimination results were not included in the source text.

7. MCC Feeder Schedule

The equipment list provides details for four motor control centre (MCC) feeders.

Feeder tag	Protective device	Rated current
04BFA-01	MCCB	250 A
04BFA-02	MCCB	250 A
04BFA-03	MCCB	160 A
04BFA-04	MCCB	125 A

Terminology

MCCB means Moulded Case Circuit Breaker. The values above are the feeder protective-device ratings supplied in the source text; they are not a substitute for checking load current, cable ampacity, fault duty, and coordination.

Scope limitation

Although the description refers to feeders 04BFA-01 through 04BFA-08, ratings for 04BFA-05 through 04BFA-08 were not provided. Those feeder details require confirmation from the complete SLD or downstream documentation.

8. Engineering Notes & References

The following notes and related document references were supplied.

Engineering notes

No.	Note
1	Protective devices coordinated as per IEC 60947.
2	Cable sizes and routes to be verified on site.
3	Refer to RAN-HHTT-0000EL-SLD-0018 for upstream HV interface.

Related documents

Document reference	Purpose / relationship
CEG-HHTT-DWG-7011	Layout reference
RAN-HHTT-0000EL-SLD-0020	Downstream MCC overview
RAN-HHTT-0000EL-SLD-0018	Upstream HV interface (referenced in note)

Field verification

Cable sizing and routing are explicitly identified as requiring on-site verification. Confirm that the referenced drawings are current and approved before using this summary for design, installation, testing, or operation.

9. Revisions, Approvals & Document Control

Revision history

Revision	Change description
Rev 03	Added MCC feeder 04BFA-04.
Rev 04	Updated incomer ACB model and fault level study results; dated 2020-04-30.

Approvals

Role	Initials / name supplied
Prepared by	GN
Checked by	DB
Approved by	RAN

Document-use statement

This 10-page PDF is a formatted expansion of the text provided by the requester. It preserves the supplied equipment data and references, adds explanatory organization, and flags information gaps without inventing missing ratings or design details. Use the approved controlled SLD and associated engineering documents as the authoritative source.

End of document